LOW-TEMPERATURE SPECIFIC HEAT CAPACITY AND LINEAR THERMAL EXPANSION OF MATERIAL FROM ASTEROID 101955 BENNU

C. P. Opeil[1], J. Biele[2], J. L. Smith[3], A. J. Ryan[4], J. L. Molaro[4], R.-L. Ballouz[5], R. J. Macke[6], H. C. Connolly, Jr.[4,7,8], D. S. Lauretta[4]

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[1]Department of Physics, Boston College, Chestnut Hill, MA, USA, [2]German Aerospace Center, Köln, Germany, [3]Los Alamos National Laboratory, Los Alamos, NM, USA, [4]Lunar & Planetary Laboratory, University of Arizona, Tucson, AZ, USA, [5]Johns Hopkins University Applied Physics Laboratory, Laurel, MD, USA, [6]Vatican Observatory, Vatican City State, [7]Department of Geology, Rowan University, Glassboro, NJ, USA, [8]Department of Earth & Planetary Sciences, American Museum of Natural History, NYC, NY, USA.

Abstract: OSIRIS-REx returned samples from asteroid (101955) Bennu on 24 September 2023. This asteroid was chosen as the target of study because of its primitive origins dating to the birth of our Solar System [1]. Bennu has a low-albedo surface and is classified as a B-type (spectrally blue) asteroid, a sub-type of the carbonaceous (C-type) asteroids. Low-temperature measurement of specific heat capacity (3–250 K), a key parameter in determining thermal inertia, allows a thermodynamic profile of this asteroid material for comparison with CM and CI meteorites and other asteroids, helping us to understand the chemical evolution of the early solar system. Thermal expansion measurements (300–12 K) provide data for thermal fatigue modeling and understanding the evolution of asteroid surface material [2,3]. Such measurements inform the primary hypothesis of the OSIRIS-REx mission of how the surface materials of Bennu have been modified by exposure to space and time.

(1) Introduction:

(a) Specific heat capacity and thermal expansion are fundamental thermodynamic and physical characteristics of asteroid material. Such data help to model various aspects of asteroid surface properties, as well as, the internal evolution of asteroids and small solar system bodies. Such characterization techniques help to identify material, composition, petrology and orbital dynamics of the asteroid itself. The Verwey transition at $T_V \sim 123.3$ K signals the presence of magnetite in the composition of Bennu's surface [4, 5].

(b) Bennu's samples were predicted to contain phyllosilicates, a type of clay, resulting from aqueous alteration of the parent body minerals. [6,7,8] Previous CM2 and CI1 meteorites studied [9] exhibit similar negative thermal expansion (NTE) as shown here in Fig. 2 & 3. Variations in the NTE shown in Fig. 2 & 3 are due to a difference in the porosity/density of a pressed powder and a naturally formed stone sample. The NTE behaviors shown here confirm the conjecture regarding the presence of phyllosilicates [8].

(c) The negative thermal expansion (NTE) shown in Fig. 2 & 3 is thought to be the result of transverse vibrations of molecules that link the layers of phyllosilicates present in the asteroid. The transverse vibrations contract the phyllosilicate layers and are most energetic in the temperature range of 200-240 K. [9] Temperatures on the surface of the asteroid Bennu fluctuate between 200 and 390 K. [10] The non-uniform NTE allow cracks to form in the asteroid's boulders. Over time we expect the comminution of boulders into regolith. [2,7,10,11,12]

(2) Specific Heat Capacity (c_p) vs. T (3-250 K)

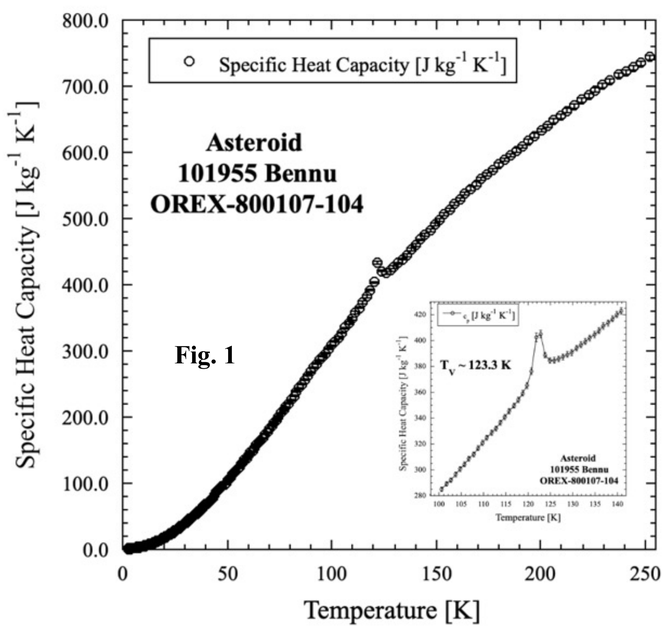


Fig. 1: Specific heat capacity vs. T for the Bennu pressed powder pellet sample [13] (OREX-800107-104) with a mass = 31.9 mg, bulk density = 2.561 g/cc. Inset: A close up of the Verwey transition, $T_V \sim 123.3$ K, confirms the presence of magnetite. [4,5]

(3) Thermal Expansion Coefficient (α) vs. T of a Pressed Sample Pellet

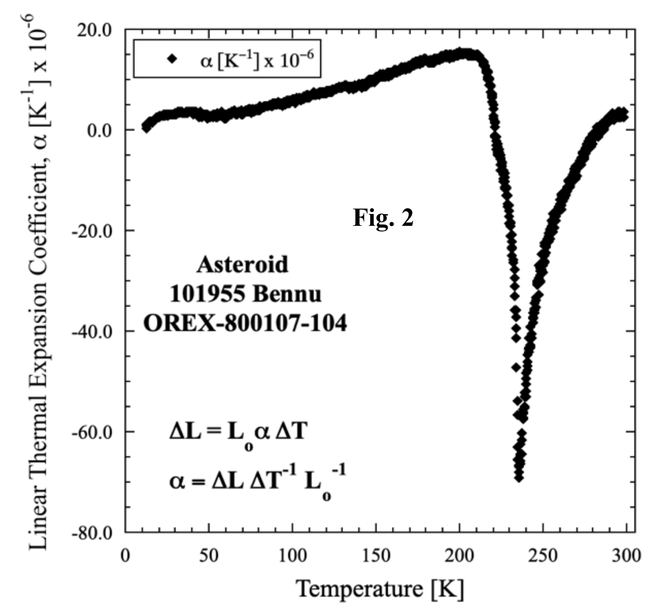


Fig. 2: Thermal expansion coefficient (α) vs. T measurements are shown for a pressed pellet sample [13] (OREX-800107-104) from T = 300-12 K. Data were taken at a cooling rate of 0.25 K/min using a Schmiedeshoff capacitive dilatometer [14] with a resolution of ~ 0.5 x 10^{-10} m.

(4) Thermal Expansion Coefficient (α) vs. T of a cut stone sample

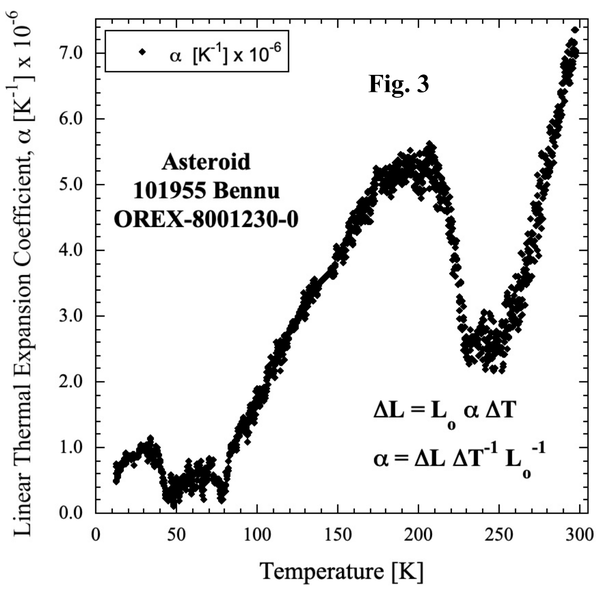


Fig. 3: Thermal expansion coefficient (α) vs. T measurements are shown for a sample cut from a single Bennu stone (OREX-8001230-0). Data were taken at a cooling rate of 0.25 K/min using a capacitive dilatometer [14] with a resolution of ~ 0.5 x 10^{-10} m.

(5) Asteroid 101955 Bennu sample pressed pellet and a cut stone sample photos

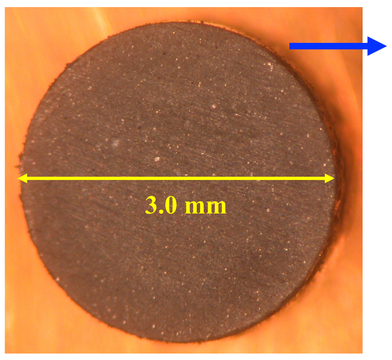
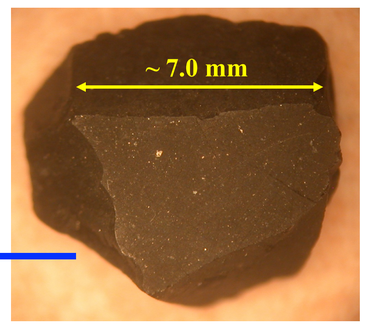


Photo 1: Pressed pellet sample (OREX-800107-104) dimensions: diameter = 3.0 mm, height = 1.761 mm, mass = 31.9 mg, & bulk density = 2.561 g/cc. The same sample (at left) was used for both specific heat capacity & linear thermal expansion data shown in Fig. 1 & 2. (Opeil photo, 11/2024).

Photo 2: Asteroid 101955 Bennu sample (OREX-800-1230-0) with mass = 0.2477 g, height = 3.578 mm was cut from a solid stone by A. Hildebrand at U. of Calgary. Linear thermal expansion data from this sample are shown in Fig. 3 (Opeil photo, 12/2024).



(6) Conclusions/Analysis:

(a) The specific heat capacity data in Fig. 1 confirm the strong low-temperature dependence and indicate the high-temperature saturation behavior at $T > 250$ K. The Verwey transition [4,5] in specific heat capacity at $T_V \sim 123.3$ K verifies the presence of magnetite as predicted by orbital spectral absorption measurements of the Bennu surface. [7, 15,16]

(b) Figures 2 & 3 demonstrate NTE behaviors of Bennu samples between 200-240 K. The predominance of phyllosilicates in Bennu surface boulders, and non-uniform surface temperature cycling between 200–350 K create near-surface diurnal thermal gradients that cause stress fractures in surface materials. We conclude from this data that over time Bennu surface boulders will fracture into regolith. [2,7,10,12,15]

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